AAI-003 | Customer Review Sentiment Analysis Report

Invertis University, Bareilly | BCA AI | Semester V | Section A

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Domain: Natural Language Processing (NLP) / Binary Text Classification
Dataset: UCI Sentiment Labelled Sentences (Dua and Graff, 2017)

1. Executive Summary & Research Objective

The objective of this project is to build, evaluate, and deploy a robust binary text classification system to categorize English customer reviews as either **Positive** or **Negative**. Utilizing public review data from Amazon, IMDb, and Yelp, we implement a complete machine learning pipeline leveraging term frequency-inverse document frequency (Sparck Jones, 1972) feature extraction combined with Logistic Regression and Multinomial Naive Bayes classifiers.

Scope Honesty: The selected benchmark dataset contains binary labels only and does not provide a Neutral class. Therefore, this implementation strictly addresses binary classification. All empirical findings and performance metrics reported herein were generated directly from execution runs on the structured data partitions.

2. Dataset Inspection and Cleaning

Review records were gathered across three domains. After parsing tab-separated text files and removing exact duplicate review entries, a total of 2,725 clean review records were retained from an initial 2,743 parsed rows.

Source Domain	Initial Parsed Rows	Clean Unique Rows	Label Balance
Amazon	1,000	993	Balanced (Positive / Negative)
IMDb	743	737	Balanced (Positive / Negative)
Yelp	1,000	995	Balanced (Positive / Negative)
Total	**2,743**	**2,725**	**Positive: 1,374 | Negative: 1,351**

Text cleaning normalized whitespace while strictly preserving original punctuation, case variations, and negation structures to retain critical sentiment signals. No conflicting labels were identified for identical review texts across sources.

3. Feature Engineering and Stratified Partitioning

To prevent vocabulary leakage and maintain rigorous experimental standards, the dataset was split into stratified partitions: **Training set (1,635 rows / 60%)**, **Validation set (545 rows / 20%)**, and **Test set (545 rows / 20%)**.

A TF-IDF vectorizer (`TfidfVectorizer`) with unigram and bigram features (`ngram_range=(1, 2)`), English stop-word removal, sublinear term frequency scaling, and a maximum feature cap of 5,000 was fitted exclusively on the training partition. Validation and test partitions were subsequently transformed using this fitted vocabulary space.

4. Model Training, Validation & Test Evaluation

Logistic Regression (configured with balanced class weights) and Multinomial Naive Bayes ($\alpha = 0.5$) were trained and evaluated. Model selection was performed on the validation partition using macro-F_1 score, where Logistic Regression achieved 0.7798 compared to 0.7650 for Naive Bayes.

Target Class	Precision	Recall	F_1 Score	Support
Negative	0.7802	0.7889	0.7845	270
Positive	0.7904	0.7818	0.7861	275
Macro Average / Total	**0.7853**	**0.7854**	**0.7853**	**545**

Overall Test Metrics: On the held-out test set, Logistic Regression attained an overall test accuracy of **0.7853**, macro precision of **0.7853**, macro recall of **0.7854**, and macro-F_1 of **0.7853**. Multinomial Naive Bayes attained an accuracy of 0.7798.

5. Sample Predictions & Operational Deployment

The trained pipeline was saved via Joblib to `model/sentiment_model_pipeline.joblib` and validated on new customer statements:

- *"This product is excellent and works exactly as expected."* → **Positive** (Confidence: 78.89%)
- *"The service was terrible and I will not return."* → **Negative** (Confidence: 85.99%)

6. References

- Sparck Jones, K. (1972). A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, 28(1), 11–21.
- Dua, D., and Graff, C. (2017). UCI Machine Learning Repository: Sentiment Labelled Sentences Data Set. UCI.